

MONTHLY ENERGY PERFORMANCE REPORT

Month: September 2026
Report period: 01 September 2026 - 30 September 2026
Generated on: 2026-09-20 14:20 UTC
Electricity rate: Rs7.00 / kWh

Executive Summary

Total Energy Used: 511.96 kWh
Average Power: 2.13 kW
Peak Power: 3.89 kW
Peak Time: 12 September 2026, 06:50 UTC
Monitored Meters: 9 / 9
Alerts: 4 Anomalies: 3 Faults: 1
High-Risk Meters: 2

What Happened This Month?

During September 2026, the system consumed 511.96 kWh of electricity. The highest recorded demand was 3.89 kW. 3 anomalies and 1 faults were detected. Two meters were classified as high maintenance risk: 2 meters classified as high maintenance risk and 7 meters classified as watch.

System Performance

Metric	Value	Explanation
Total Energy Used	511.96 kWh	Total electrical energy cons
Average Power	2.13 kW	Mean power draw over the rep
Peak Power	3.89 kW	Highest instantaneous power
Peak Time	12 September 2026, 0	Timestamp of the peak demand
Active Meters	9 / 9	PZEM meters reporting data o

Meter-wise Energy Consumption

Meter	Energy Used (kWh)		Peak Power (kW)		Anomalies	Faults	Maintenance Risk	Forecast	Bill Status
PZEM 1	48.42	0.42	0	0	Watch		FORECAST	OK	
PZEM 2	48.25	0.42	0	0	Watch		FORECAST	OK	
PZEM 3	48.40	0.41	1	0	Watch		FORECAST	OK	
PZEM 4	86.84	0.74	0	0	High maintenance r		FORECAST	OK	
PZEM 5	48.26	0.40	0	1	Watch		FORECAST	OK	
PZEM 6	48.22	0.40	1	0	Watch		FORECAST	OK	
PZEM 7	48.20	0.41	0	0	Watch		FORECAST	OK	
PZEM 8	87.04	0.71	0	0	High maintenance r		FORECAST	OK	
PZEM 9	48.33	0.41	1	0	Watch		FORECAST	OK	

Peak Demand Analysis

Maximum system power: 0.74 kW
Peak date: 16 September 2026
Peak time: 06:30 UTC
Peak above baseline: 0.36 kW
Recurring high-demand periods are visible in the charts below.

Alerts & Abnormal Events

Summary: 3 anomalies, 1 faults, 0 emergencies

PZEM 5 - Over-voltage condition detected

Date: 19 September 2026

Status: Warning

Maintenance Risk

- 2 meters: High maintenance risk

- 7 meters: Watch

High-risk meters: PZEM 4, PZEM 8

Watch meters: PZEM 1, PZEM 2, PZEM 3, PZEM 5, PZEM 6, PZEM 7, PZEM 9

Energy-Saving Recommendations

PZEM 1 - Predictable High-Usage Period

Priority: Medium

Observation: Forecast shows a predictable high-load window 06:30-09:00 (median 368 W vs typical 204 W).

PZEM 1 - Consider Shifting Non-Critical Loads

Priority: Medium

Observation: Recurring high-demand window 05:00-07:30 (median 349 W vs typical 210 W).

PZEM 2 - Predictable High-Usage Period

Priority: Medium

Observation: Forecast shows a predictable high-load window 05:30-08:00 (median 365 W vs typical 213 W).

PZEM 2 - Consider Shifting Non-Critical Loads

Priority: Medium

Observation: Recurring high-demand window 05:30-08:00 (median 353 W vs typical 207 W).

PZEM 3 - Predictable High-Usage Period

Priority: Medium

Observation: Forecast shows a predictable high-load window 05:30-08:00 (median 383 W vs typical 215 W).

PZEM 3 - Consider Shifting Non-Critical Loads

Priority: Medium

Observation: Recurring high-demand window 05:30-08:00 (median 351 W vs typical 211 W).

PZEM 4 - Reduce Unnecessary Standby Consumption

Priority: Medium

Observation: Minimum observed power (idle/standby baseline) is 79 W, indicating an always-on load. Curtailing ~30% of it is a realistic saving.

Estimated saving: 3.88 kWh

Estimated cost impact: Rs27.18

PZEM 4 - Predictable High-Usage Period

Priority: Medium

Observation: Forecast shows a predictable high-load window 04:30-07:00 (median 639 W vs typical 371 W).

PZEM 4 - Consider Shifting Non-Critical Loads

Priority: Medium

Observation: Recurring high-demand window 04:30-07:00 (median 633 W vs typical 372 W).

PZEM 5 - Predictable High-Usage Period

Priority: Medium

Observation: Forecast shows a predictable high-load window 06:30-08:30 (median 355 W vs typical 209 W).

PZEM 5 - Consider Shifting Non-Critical Loads

Priority: Medium

Observation: Recurring high-demand window 06:00-08:30 (median 350 W vs typical 208 W).

PZEM 6 - Predictable High-Usage Period

Priority: Medium

Observation: Forecast shows a predictable high-load window 05:30-08:00 (median 366 W vs typical 212 W).

PZEM 6 - Consider Shifting Non-Critical Loads

Priority: Medium

Observation: Recurring high-demand window 04:30-07:00 (median 349 W vs typical 208 W).

PZEM 7 - Predictable High-Usage Period

Priority: Medium

Observation: Forecast shows a predictable high-load window 05:00-07:30 (median 374 W vs typical 202 W).

PZEM 7 - Consider Shifting Non-Critical Loads

Priority: Medium

Observation: Recurring high-demand window 06:00-08:30 (median 355 W vs typical 208 W).

PZEM 8 - Reduce Unnecessary Standby Consumption

Priority: Medium

Observation: Minimum observed power (idle/standby baseline) is 80 W, indicating an always-on load. Curtailing ~30% of it is a realistic saving.

Estimated saving: 4.10 kWh

Estimated cost impact: Rs28.72

PZEM 8 - Predictable High-Usage Period

Priority: Medium

Observation: Forecast shows a predictable high-load window 06:30-09:00 (median 666 W vs typical 374 W).

PZEM 8 - Consider Shifting Non-Critical Loads

Priority: Medium

Observation: Recurring high-demand window 05:00-07:30 (median 634 W vs typical 376 W).

PZEM 9 - Predictable High-Usage Period

Priority: Medium

Observation: Forecast shows a predictable high-load window 05:00-07:30 (median 363 W vs typical 211 W).

PZEM 9 - Consider Shifting Non-Critical Loads

Priority: Medium

Observation: Recurring high-demand window 05:00-07:30 (median 352 W vs typical 208 W).

SYSTEM - Consider Shifting Non-Critical Loads

Priority: Medium

Observation: Recurring high-demand window 05:00-07:30 (median 3718 W vs typical 2235 W).

PZEM 1 - Reduce Unnecessary Standby Consumption

Priority: Low

Observation: Minimum observed power (idle/standby baseline) is 44 W, indicating an always-on load. Curtailing ~30% of it is a realistic saving.

Estimated saving: 2.17 kWh

Estimated cost impact: Rs15.18

PZEM 2 - Reduce Unnecessary Standby Consumption

Priority: Low

Observation: Minimum observed power (idle/standby baseline) is 45 W, indicating an always-on load. Curtailing ~30% of it is a realistic saving.

Estimated saving: 2.34 kWh

Estimated cost impact: Rs16.35

PZEM 3 - Reduce Unnecessary Standby Consumption

Priority: Low

Observation: Minimum observed power (idle/standby baseline) is 44 W, indicating an always-on load. Curtailing ~30% of it is a realistic saving.

Estimated saving: 2.21 kWh

Estimated cost impact: Rs15.45

PZEM 5 - Reduce Unnecessary Standby Consumption

Priority: Low

Observation: Minimum observed power (idle/standby baseline) is 43 W, indicating an always-on load. Curtailing ~30% of it is a realistic saving.

Estimated saving: 2.09 kWh

Estimated cost impact: Rs14.60

PZEM 6 - Reduce Unnecessary Standby Consumption

Priority: Low

Observation: Minimum observed power (idle/standby baseline) is 45 W, indicating an always-on load. Curtailing ~30% of it is a realistic saving.

Estimated saving: 2.34 kWh

Estimated cost impact: Rs16.35

PZEM 7 - Reduce Unnecessary Standby Consumption

Priority: Low

Observation: Minimum observed power (idle/standby baseline) is 43 W, indicating an always-on load. Curtailing ~30% of it is a realistic saving.
Estimated saving: 2.09 kWh
Estimated cost impact: Rs14.65

PZEM 9 - Reduce Unnecessary Standby Consumption

Priority: Low
Observation: Minimum observed power (idle/standby baseline) is 45 W, indicating an always-on load. Curtailing ~30% of it is a realistic saving.
Estimated saving: 2.28 kWh
Estimated cost impact: Rs15.95

Energy Cost

Electricity rate: Rs7.00 / kWh
Energy consumed: 511.96 kWh
Estimated energy cost: Rs9450.00
Estimated total energy: 1350.00 kWh

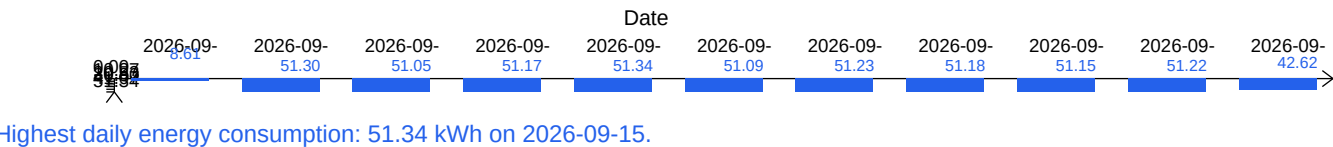
Energy Forecast

Forecast unavailable
Insufficient forecast data was available for the system during this report period.

Charts

Daily Energy Consumption

Energy (kWh)



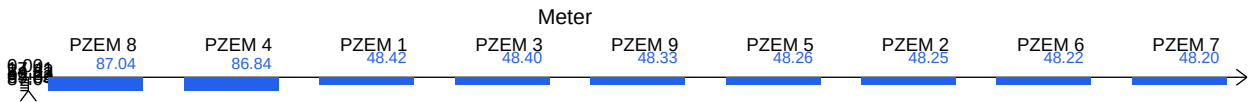
Daily Peak Power

Peak Power (kW)

Highest daily peak: 3.89 kW on 2026-09-12.
Key value: 2026-09-12 = 3.89 kW

Energy Consumption by Meter

Energy (kWh)



Highest daily energy consumption: 87.04 kWh on PZEM 8.

Top Energy Consumers:

Meter	Energy
PZEM 8	87.04 kWh
PZEM 4	86.84 kWh
PZEM 1	48.42 kWh
PZEM 3	48.40 kWh
PZEM 9	48.33 kWh

Anomalies and Faults by Meter

Count



Technical Details / Appendix

Internal recommendation codes (traceability):

RESPOND_PREDICTABLE_HIGH_LOAD - PZEM 1 - Medium
SHIFT_NON_CRITICAL_LOAD - PZEM 1 - Medium
RESPOND_PREDICTABLE_HIGH_LOAD - PZEM 2 - Medium
SHIFT_NON_CRITICAL_LOAD - PZEM 2 - Medium
RESPOND_PREDICTABLE_HIGH_LOAD - PZEM 3 - Medium
SHIFT_NON_CRITICAL_LOAD - PZEM 3 - Medium
REDUCE_IDLE_CONSUMPTION - PZEM 4 - Medium
RESPOND_PREDICTABLE_HIGH_LOAD - PZEM 4 - Medium
SHIFT_NON_CRITICAL_LOAD - PZEM 4 - Medium
RESPOND_PREDICTABLE_HIGH_LOAD - PZEM 5 - Medium
SHIFT_NON_CRITICAL_LOAD - PZEM 5 - Medium
RESPOND_PREDICTABLE_HIGH_LOAD - PZEM 6 - Medium
SHIFT_NON_CRITICAL_LOAD - PZEM 6 - Medium
RESPOND_PREDICTABLE_HIGH_LOAD - PZEM 7 - Medium
SHIFT_NON_CRITICAL_LOAD - PZEM 7 - Medium
REDUCE_IDLE_CONSUMPTION - PZEM 8 - Medium
RESPOND_PREDICTABLE_HIGH_LOAD - PZEM 8 - Medium
SHIFT_NON_CRITICAL_LOAD - PZEM 8 - Medium
RESPOND_PREDICTABLE_HIGH_LOAD - PZEM 9 - Medium
SHIFT_NON_CRITICAL_LOAD - PZEM 9 - Medium
SHIFT_NON_CRITICAL_LOAD - SYSTEM - Medium
REDUCE_IDLE_CONSUMPTION - PZEM 1 - Low
REDUCE_IDLE_CONSUMPTION - PZEM 2 - Low
REDUCE_IDLE_CONSUMPTION - PZEM 3 - Low
REDUCE_IDLE_CONSUMPTION - PZEM 5 - Low
REDUCE_IDLE_CONSUMPTION - PZEM 6 - Low
REDUCE_IDLE_CONSUMPTION - PZEM 7 - Low
REDUCE_IDLE_CONSUMPTION - PZEM 9 - Low

Fault types (internal codes):

OVER_VOLTAGE - PZEM 5 - WARNING

Report generated by Smart Energy Monitoring System.